SMART INDIA HACKATHON 2026

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- **Problem Statement ID** – SIH26026
- **Problem Statement Title** – Development of Mobile (Quadruped)/Handheld Device/System for Real-Time Detection of Narcotics and Explosives across Indian Railways.
- **Theme** – Blockchain & Cybersecurity
- **PS Category** – Hardware
- **Team ID** – 133660
- **Team Name** – ABHEDYA: The UnBreachable Shield

ABHEDYA: The UnBreachable Shield

ABHEDYA is a **portable, non-contact Raman spectroscopy** based screening system designed for **rapid** detection of suspicious narcotic and explosive materials in railway environments. It provides a **low cost**, accessible alternative for first level molecular screening, using a **OpenRAMAN** open source reference design of Raman sensing **unit** and Raspberry Pi for spectral acquisition and processing. **PCA and machine learning classification** compare measured **Raman molecular fingerprints** with a reference spectral library. The approach achieves the maximum practical spectroscopic capability within a **minimal cost architecture**, while enabling future integration with **certified confirmation techniques.**

EXISTING PROBLEM	ABHEDYA'S RESPONSE
Expensive analytical equipment	ABHEDYA's **low-cost, compact Raman architecture brings molecular-level screening** closer to the point of inspection.
Manual inspection	ABHEDYA enables **non-contact optical measurement, allowing preliminary screening** without physically handling the suspected substance.
Environmental variability and interference	ABHEDYA uses controlled **optical acquisition, background subtraction, preprocessing, and reference libraries** to help separate useful spectral information from interference.
Limited physical coverage	ABHEDYA provides **handheld + quadruped deployment, allowing the same Raman sensing and ML pipeline** to be used across different inspection scenarios.
Lack of traceability	ABHEDYA **digitally logs** the spectrum, classification, confidence, timestamp, and measurement metadata, creating an auditable screening record.

Existing Approach	Limitation	ABHEDYA's Approach
Laboratory spectroscopy	Expensive and centralized	Low-cost portable Raman architecture
Commercial Raman	Higher acquisition cost	Exclusively developed Raman prototype from open source OpenRAMAN.
Visual inspection	No molecular information	Molecular Fingerprint Classification
Simple sensor thresholds	Limited Vapor pressure information	Spectrum and chemometrics/ML
Forced binary classification	Poor readings may be misclassified	Three principle verification
Fixed systems	Limited physical coverage	Handheld & quadruped deployment
Manual records	Limited Measurement Traceability	Digital metadata logging
Screening treated as final result	Risk of overclaiming	ESCALATE to certified confirmation

Principle of Raman Effect in ABHEDYA

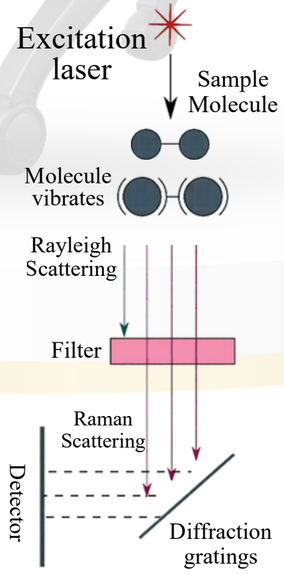


Fig. 01 Principle of Raman Effect

ABHEDYA uses the Raman effect to **obtain a molecular fingerprint** of a suspicious material **without physically touching it**.

A 785 nm laser illuminates the material. Light that undergoes Rayleigh scattering is rejected using an **edge/long-pass filter**. A portion undergoes **Raman scattering**, producing wavelength shifts that correspond to the material's molecular vibrations.

The Raman-shifted light is collected by the optics and passed through the **spectrometer**, which converts it into a **Raman spectrum.** The Raspberry Pi then **performs dark correction, baseline correction, noise reduction and feature extraction, and compares the spectrum** with the reference spectral library **using PCA + ML analysis.**

Website | Detailed Workflow | Explanation Video

Cad Module | Report | ML model Git Repo

TECHNICAL APPROACH

ABHEDYA is a **Raman first**, **SURFACE BASED** edge **AI screening** system integrating **molecular fingerprinting**, **contactless** sensing, **Machine Learning based** analysis, and **three phase decisions**. Designed for handheld and quadruped deployment, it offers a **low-cost, portable** and **highly problem aligned approach** for this problem statement.

Optical Component Layout

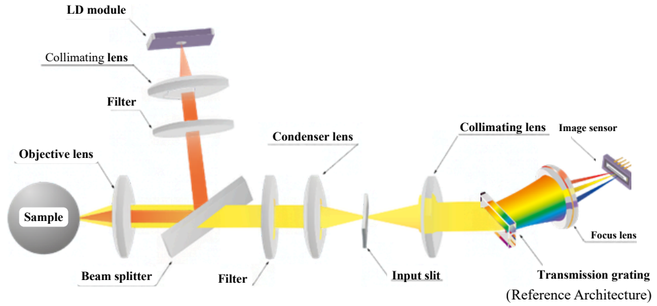


Fig. 02 Optical Layout of ABHEDYA optical component

Block diagram

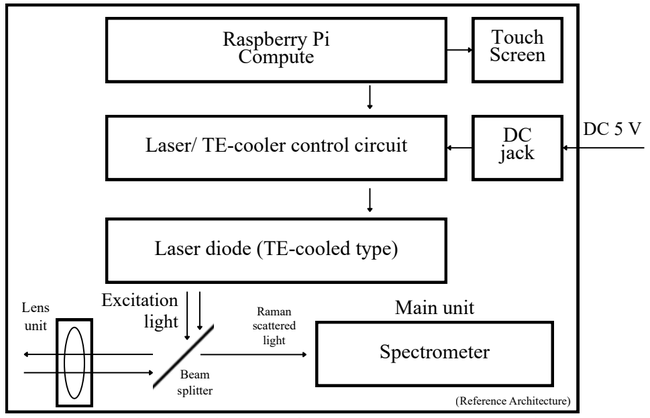


Fig. 03 Block Daigram of ABHEDYA's Approach

ABHEDYA Workflow

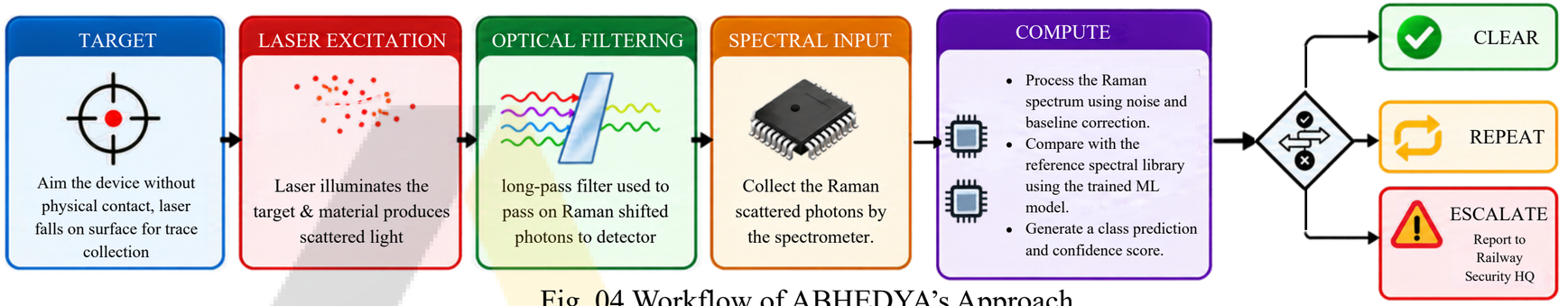


Fig. 04 Workflow of ABHEDYA's Approach

Experimental & Practical proof of Idea

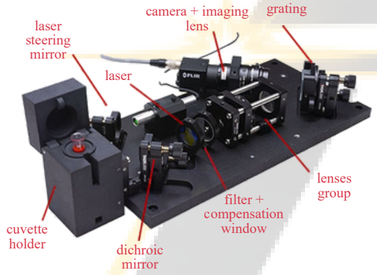


Fig. 05 base spectrometer with standard liquid cuvette

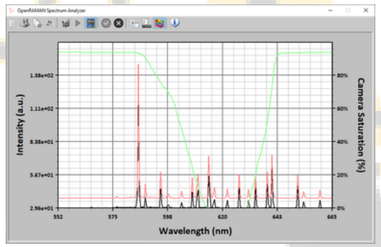
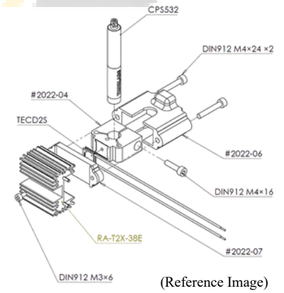


Fig. 06 OpenRAMAN Spectrum Analyzer



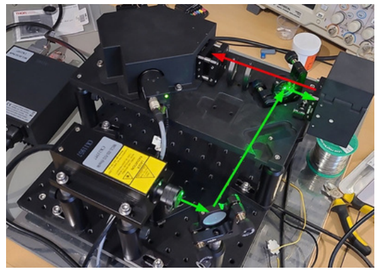


Fig. 07 TEC laser module Thorlabs CPS532

ABHEDYA's Development Plan Model view



(REFRENCES IMAGE)



(REFRENCES IMAGES)

Do visit and review our ABHEDYA

- Website
- Report
- Cad Module
- Explanation Video
- Detailed Workflow
- ML model Git Repo

FEASIBILITY AND VIABILITY

Feasibility of ABHEDYA

Area	Feasibility
Detection	ABHEDYA provides 785 nm Raman spectroscopic **molecular fingerprints** from materials without physical contact.
Compact hardware	ABHEDYA is single **portable unit and Mobile Qudruped.**
Edge AI	ABHEDYA performs **spectral preprocessing, PCA and lightweight ML classification** locally.
Deployment	ABHEDYA's core can be integrated into a **handheld device or quadruped** platform.
Cost	ABHEDYA's **custom compact architecture** significantly **reduces system cost compared** with laboratory Raman instruments.
Scalability	ABHEDYA **continuously improves** its spectral library and ML model by validated **real world field data** collected during deployment, **enabling periodic retraining and adaptation to new materials** and operating conditions.

"Our literature review did not identify a study demonstrating this specific low-cost, contactless Raman combination for real-world railway deployment."

Challenges faced and ABHEDYA approach evolution during Ideation Period

Approach	Why we Considered	Key Limitation	ABHEDYA's Mitigation
Contact-based Testing	Simple presumptive testing	Physical contact, consumables and sample handling	Non-contact Raman screening
LFA / Immunoassay Strips	Target-specific presumptive detection	Consumables, less shelf life, recurring cost and physical contact	Reusable optical sensing architecture
Gas Sensor Array	Low-cost, compact embedded sensing	Low vapour pressure of many narcotics; diesel/perfume interference	Molecular spectral sensing instead of vapour - dependent detection
ATR-FTIR	Chemically informative molecular spectra	Contact/sample constraints	Non-contact Raman workflow
AS7265x Multispectral Sensor	Low-cost, compact, contactless sensing	Broad spectral signatures are less molecularly specific	Raman molecular fingerprinting
Field Deployment	Real-world screening capability	Changing environmental and sample conditions	Calibration checks, REPEAT state periodic model/library updates
Safety & High-stakes Decisions	Railway security application	Laser safety and consequences of incorrect classification	Enclosed optical path, interlock, three phase detection secondary certified testing

Viability of ABHEDYA

Technical viability	ABHEDYA Uses **Raman spectroscopy principles combined with compact optics, Raspberry Pi** processing and lightweight ML.	**Maintenance viability**	ABHEDYA uses Optical hardware that **avoids recurring consumables associated with strip-based testing**; calibration and software updates can be performed periodically **autonomously**.
Operational viability	ABHEDYA uses **Non-contact screening** which reduces the need for personnel to directly handle suspicious materials during initial screening.	**Deployment viability**	ABHEDYA has Raspberry Pi-based **edge processing** which allows spectral analysis and preliminary decisions directly on the device.
Economic viability	ABHEDYA has **custom compact Raman architecture** can reduce dependence on expensive laboratory-grade instruments and supports reusable sensing.	**Decision viability**	ABHEDYA has **three phase decision** that separates acceptable, ambiguous and suspicious results rather than forcing every measurement into a binary decision.
AI & Data viability	The spectral library and ML model used in ABHEDYA **continuously improves** using validated real-world field data collected during deployment.	**Future scalability**	Additional validated spectral classes, improved models and potentially enhanced Raman techniques such as SERS can be incorporated later in ABHEDYA.

IMPACT AND BENEFITS

Impact from ABHEDYA



Railway Security Personnel

Enables rapid, non-contact preliminary screening of suspicious materials.



RPF / Railway Authorities

Supports faster decision-making at checkpoints and during inspections.



Passengers & Public

Contributes to safer railway environments by enabling earlier identification of suspicious substances.



Security in Hazardous Situations

Handheld and quadruped deployment can extend inspection capability while reducing the need for immediate close approach.



Future Deployment

Scalable spectral libraries allow the system to adapt to additional validated material classes.

Benefits of ABHEDYA



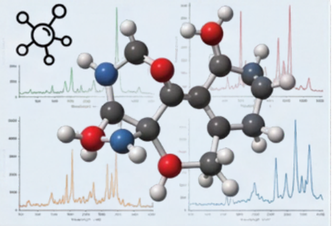
Non-contact screening:

Reduces direct handling of suspicious materials during initial screening.



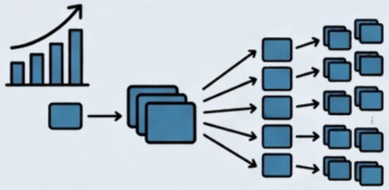
Rapid analysis:

Raman spectrum can be acquired and processed at the edge (near real-time).



Molecular fingerprinting:

Provides chemically informative spectral signatures rather than relying only on vapours or broad



Continuously improvable

Validated field data can expand the reference library and improve the ML model over time.



Scalable

Additional validated spectral classes can be incorporated into the system.



Lower youth drug addiction

Helps reduce availability and trafficking of narcotics, contributing to a healthier and safer youth population.



More secure India

Strengthens railway security and national security by detecting narcotics and explosives earlier in the supply chain.



Pre-incident detection

Enables earlier identification of suspicious materials, helping prevent potential incidents before they occur.



Safer society

Contributes to a safer railway environment, protects lives and property, and supports a more secure and resilient India.



Reusable

Uncertainty-aware

CLEAR / REPEAT / ESCALATE avoids forcing ambiguous measurements into a binary decision.

CLEAR	REPEAT	ESCALATE
No Threat Detected	Scan Again	Suspicious Material

(Here, Images are used for Visiual Cliarity in this slide)

RESEARCH AND REFERENCES

Objective

To understand real-world security challenges, study existing detection methods, and gain hands-on knowledge of Raman spectroscopy to develop ABHEDYA.

Key Findings

1. High passenger volume increases screening challenges
2. Manual inspection is time-consuming and inconsistent
3. Need for non-contact, portable, real-time detection
4. Raman spectroscopy provides unique molecular fingerprint
5. Potential for field-deployable, low-cost solution

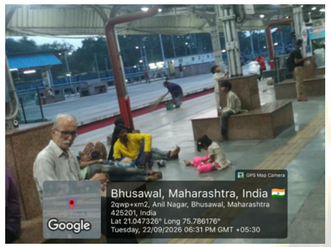
Outcomes

1. Validated real-world need for ABHEDYA
2. Understood practical constraints
3. Gained hands-on knowledge of Raman spectroscopy
4. Identified research gaps and opportunities
5. Defined technical approach for prototype

Ground-Zero Field & Laboratory Research

















- **Pune Junction** — Studied real-world railway security conditions and existing screening practices.
- **Nagpur Junction** — Observed passenger movement, operational challenges, and security requirements.
- **Bhusaval Junction** — Gained practical understanding of on-ground inspection and screening challenges.
- **SPPU Raman Spectroscopy Laboratory** — Gained hands-on experience with laboratory-based Raman spectroscopy.
- **Mumbai CSMT - Solapur Vande Bharat Express (Train No. 22225)** — Conducted an on-train field study to understand security challenges inside

References & Literature Survey

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